Name of Course	**Bachelor of Computer Application (BCA)**
Semester	**V**
Name of Subject	**Computer Network**
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Unit I

1. Introduction to computer network

1.1 What is Network? , Benefits of Networking

1.2 Wired Transmission media – Magnetic media, Twisted Pair, Coaxial Cable, Fiber Optics

1.3 Wireless Transmission media – The Electromagnetic Spectrum, Radio Transmission, Microwave Transmission, Infrared

1.4 Topologies with advantages & disadvantages:-Bus, Ring, Star, Tree, Mesh

1.5 Types of Networks – LAN, MAN, WAN

1.1 What is Network? , Benefits of Networking:

What is a Network?

A **network** is a group of two or more **interconnected devices or systems** that can share data, resources, and information. These devices communicate using **communication protocols** over wired or wireless mediums. Common examples include **computer networks**, **mobile networks**, and the **Internet**.

Benefits of Networking:

1. Resource Sharing:

Devices like printers, scanners, and files can be shared among multiple users, reducing cost and improving efficiency.

2. Data Sharing and Communication:

Users can easily exchange information through emails, messaging, or file transfers.

3. Centralized Data Management:

Data can be stored, backed up, and managed from a central server, making access and maintenance easier.

4. Improved Security and Access Control:

Networks allow administrators to manage user permissions, apply security policies, and protect sensitive data.

5. Remote Access and Collaboration:

Users can access network resources and collaborate with others from different locations in real-time.

In summary, a network connects devices to enable communication, collaboration, and resource sharing, making it essential in modern computing environments.

1.2 Wired Transmission media – Magnetic media, Twisted Pair, Coaxial Cable, Fiber Optics:

Wired Transmission Media:

Wired transmission media refers to the **physical cables or wires** used to transmit data signals from one device to another.

1. Magnetic Media:

It includes **hard disks, floppy disks, and magnetic tapes**.
Data is stored magnetically and transferred by physically moving the media between devices.
Mostly used for **offline data transfer and backup**.

2. Twisted Pair Cable:

Made of **two insulated copper wires twisted together**.
Commonly used in **telephone lines and LANs (e.g., Ethernet cables)**.
Types: **Unshielded (UTP)** and **Shielded (STP)**.
Low cost and easy to install but has limited bandwidth and more signal interference.

3. Coaxial Cable:

Consists of a **central copper conductor**, **insulating layer**, **metallic shield**, and **outer cover**.
Provides **better shielding** than twisted pair.
Used in **cable TV networks and broadband internet**.

4. Fiber Optic Cable:

Transmits data as **light signals** through **glass or plastic fibers**.
Offers **high speed, high bandwidth, and long-distance transmission**.
Immune to electromagnetic interference.
Used in **high-speed internet, telecommunication, and backbone networks**.

In summary, each wired media type has different applications based on speed, distance, and cost. Fiber optics offer the fastest and most reliable transmission, while twisted pair is cost-effective for short distances.

1.3 Wireless Transmission media – The Electromagnetic Spectrum, Radio Transmission, Microwave Transmission, Infrared

Wireless Transmission Media:

Wireless transmission media use **electromagnetic waves** to transmit data **without physical wires**. It enables **remote communication** between devices over short or long distances.

1. The Electromagnetic Spectrum:

It refers to the **range of all types of electromagnetic radiation**, including **radio waves, microwaves, infrared, visible light, etc.**

Different parts of the spectrum are used for **different types of wireless communication** depending on frequency and wavelength.

2. Radio Transmission:

Uses **low-frequency radio waves** to transmit signals.
Can **penetrate walls** and travel long distances.
Used in **AM/FM radios, mobile phones, and wireless networking (Wi-Fi)**.
Offers **omnidirectional** communication.

3. Microwave Transmission:

Uses **high-frequency radio waves (1–100 GHz)**.
Requires **line-of-sight** between transmitter and receiver.
Used for **television signals, satellite communication, and long-distance telephone calls**.
Has higher bandwidth than radio waves.

4. Infrared (IR):

Uses **infrared light waves** for very short-range communication.
Cannot pass through walls.
Commonly used in **TV remote controls, short-range data transfer (like file sharing between phones)**.

In summary, wireless media allow flexible and mobile communication using different parts of the electromagnetic spectrum, each suited for specific distances, data rates, and applications.

1.4 Topologies with advantages & disadvantages:-Bus, Ring, Star, Tree, Mesh:

1. Bus Topology:

Structure:

All devices are connected to a single central cable (backbone).

Advantages:

Easy to implement and cost-effective.
Requires less cable than other topologies.

Disadvantages:

Difficult to troubleshoot.
A break in the backbone affects the entire network.
Limited cable length and number of devices.

2. Ring Topology:

Structure:

Devices are connected in a circular fashion, where each device is connected to two others.

Advantages:

Data flows in one direction, reducing data collisions.
Good performance under light loads.

Disadvantages:

Failure of one node can affect the entire network.
Difficult to reconfigure or add new devices.

3. Star Topology:

Structure:

All devices are connected to a central hub or switch.

Advantages:

Easy to install and manage.
Failure in one device doesn't affect others.

Disadvantages:

If the central hub fails, the entire network goes down.
Requires more cable than bus or ring.

4. Tree Topology:

Structure:

A hybrid of star and bus; groups of star-configured networks are connected to a main backbone.

Advantages:

Scalable and easy to manage.
Fault isolation is easier.

Disadvantages:

Backbone failure can bring down the whole system.
Requires more cable and configuration.

5. Mesh Topology:

Structure:

Every device is connected to every other device.

Advantages:

Provides high redundancy and reliability.
Failure of one link does not affect communication.

Disadvantages:

Very costly due to a large number of cables and ports.
Complex to install and maintain.

In summary, each topology suits different needs based on **cost, scalability, fault tolerance**, and **ease of maintenance**.

1.5 Types of Networks – LAN, MAN, WAN:

1. LAN (Local Area Network):

A network that connects **computers within a small geographical area** such as a home, school, or office.
High data transfer speed and **low cost**.
Example: Connecting PCs and printers in a computer lab.

Advantages:

Fast and secure communication.
Easy to share resources like files and printers.

Disadvantages:

Limited to a small area.
Maintenance may be needed if the number of users grows.

2. MAN (Metropolitan Area Network):

Covers a **larger area than LAN**, like a **city or large campus**.
Connects multiple LANs using routers or fiber optic cables.
Example: A network connecting branches of a bank across a city.

Advantages:

Provides high-speed communication across a city.
Useful for organizations with multiple buildings or offices.

Disadvantages:

More expensive than LAN.
Needs skilled management and maintenance.

3. WAN (Wide Area Network):

Covers a **very large area**, such as **countries or continents**.
The **Internet** is the best example of a WAN.
Uses public or leased communication lines like satellites or undersea cables.

Advantages:

Enables long-distance communication and global connectivity.
Supports large-scale business operations.

Disadvantages:

Expensive to set up and maintain.
Lower speed and more security risks compared to LAN/MAN.

In summary, the type of network—**LAN, MAN, or WAN—**depends on the geographical area, number of devices, and communication needs.